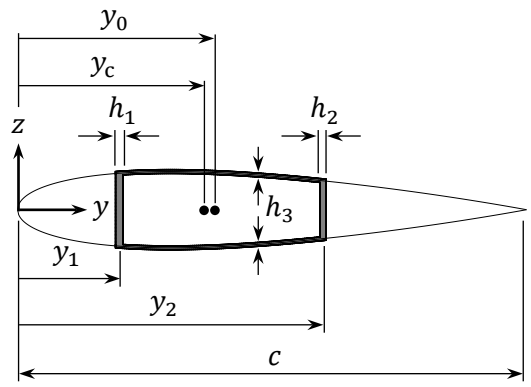
COMPUTATIONAL ENGINEERING

Project – Wing modelling

The objective of this assignment is to validate the beam and shell models studying the bending and torsion response of the wingbox structure depicted in Figure 1. The material considered has a Young's modulus of $E = 75$ GPa, a Poisson's ratio of $\nu = 0.3$, and a density of $\rho = 2200$ kg/m^3. The geometric and section properties of the wingbox are given in Table 1 below.

Table 1. Geometric properties of the wingbox.

NACA0015 profile	Data	Value	Property	Value
	c (m)	0.85	Area A (m^2)	0.0246
	b (m)	3	Inertia $I_{y'}$ (m^4)	0.727 x 10^{-4}
	y_0 (m)	0.294	Inertia $I_{z'}$ (m^4)	4.002 x 10^{-4}
	y_1 (m)	0.125	Polar inertia J (m^4)	4.729 x 10^{-4}
	y_2 (m)	0.475	Shear correction $k_{y'}$	0.6873
	h_1 (mm)	35	Shear correction $k_{z'}$	0.1843
	h_2 (mm)	30	Torsion constant k_t	0.4602
	h_3 (mm)	25	Shear center y_c (m)	0.285

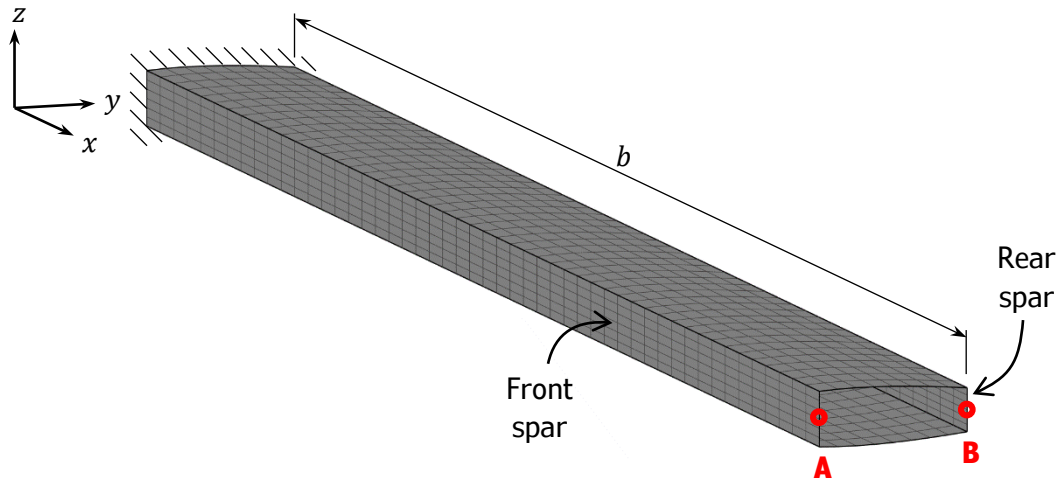


Figure 1. Wingbox structure.

Part I – Beam test

1) Model the wingbox structure with beam elements using the data given in **`beam.mat`** file and obtain the corresponding stiffness and mass matrices.
2) Perform a bending test resulting from applying a vertical shear load $F_z = 1$ N at the shear center of the wing tip and compute:
   a) Vertical deflection distribution along the wingspan, u_z vs. x.
   b) Twist angle distribution along the wingspan, θ_x vs. x.
3) Perform a torsion test resulting from applying a torque along the x-axis $M_x = 1$ N·m at the wing tip and compute:
   a) Vertical deflection distribution along the wingspan, u_z vs. x.
   b) Twist angle distribution along the wingspan, θ_x vs. x.
4) Compute the first 6 natural modes and frequencies and, for each one, represent the deflection and twist (u_y, u_z, θ_x) along the wingspan.

Note: The mesh variables contained in **`beam.mat`** file are: **`xn`** (nodal coordinates matrix), **`Tn`** (nodal connectivities matrix), and **`Tm`** (material connectivities matrix). You can use the script file **`MAIN_beam.m`** as template to develop the code to get the requested results.

Note: For these tests, do **not** take into account the weight of the structure.

Part II – Shell test

1) Model the wingbox structure with flat shell elements using the data given in **`shell.mat`** file and obtain the corresponding stiffness and mass matrices.
2) Perform a bending test resulting from applying a vertical shear load $F_z = 1$ N at the shear center of the wing tip and compute:
   a) Average vertical deflection distribution along the wingspan, $\bar{u}_z$ vs. x.
   b) Average twist angle distribution along the wingspan, $\bar{\theta}_x$ vs. x.

   Hint: To model the application of a shear load F_z at the shear center, apply the following equivalent **point vertical forces** on points A and B of the wing tip:

$$F_{\mathrm{A}} = F_z \frac{y_2 - y_{\mathrm{c}}}{y_2 - y_1}; \quad F_{\mathrm{B}} = F_z \frac{y_{\mathrm{c}} - y_1}{y_2 - y_1}$$

3) Perform a torsion test resulting from applying a torque along the x-axis $M_x = 1$ N·m at the wing tip and compute:
   a) Average vertical deflection distribution along the wingspan, $\bar{u}_z$ vs. x.
   b) Average twist angle distribution along the wingspan, $\bar{\theta}_x$ vs. x.

   Hint: To model the application of a torsion moment M_x, apply the following equivalent **point vertical forces** on points A and B of the wing tip:

$$F_{\mathrm{A}} = -\frac{M_x}{y_2 - y_1}; \quad F_{\mathrm{B}} = \frac{M_x}{y_2 - y_1}$$

4) Compute the first 6 natural modes and frequencies and, for each one, represent the average deflection and twist $(\bar{u}_y, \bar{u}_z, \bar{\theta}_x)$ along the wingspan.

Hint: The **average** deflection and twist angles can be derived from the corresponding displacements $u_y^{(i)}$ and $u_z^{(i)}$ obtained in the front spar ($i = 1$) and rear spar ($i = 2$) centerlines, according to these expressions:

$$\bar{\theta}_x = \frac{u_z^{(2)} - u_z^{(1)}}{y_2 - y_1}; \quad \bar{u}_z = u_z^{(1)} + \bar{\theta}_x(y_c - y_1); \quad \bar{u}_y = \frac{u_y^{(1)} + u_y^{(2)}}{2}$$

Hint: To help apply the boundary condition (clamped root), the indices of the nodes contained in the wingbox root are stored in the **`indRoot`** array.

Note: The mesh variables contained in **`shell.mat`** files are: **`xn`** (nodal coordinates matrix), **`Tn`** (nodal connectivities matrix), and **`Tm`** (material connectivities matrix), **`indRoot`** (array containing indices of nodes in the root section), **`indPointA`** (index of node corresponding to point A at the wing tip), **`indPointB`** (index of node corresponding to point B at the wing tip), **`indSpar1`** (array containing indices of nodes in front spar centerline), **`indSpar2`** (array containing indices of nodes in rear spar centerline). The material index 1 refers to the front spar, 2 refers to the rear spar, and 3 to the upper and lower surfaces of the wingbox. You can use the script file **`MAIN_shell.m`** as template to develop the code to get the requested results.

Note: For these tests, do **not** take into account the weight of the structure.

Part III – Wing structure

The same wingbox structure is now reinforced with **ribs** of thickness $h_r = 15$ mm, **stringers** of circular cross-section of diameter $D = 40$ mm, and covered by a **skin** of thickness $h_s = 10$ mm to offer aerodynamic capabilities to the wing.

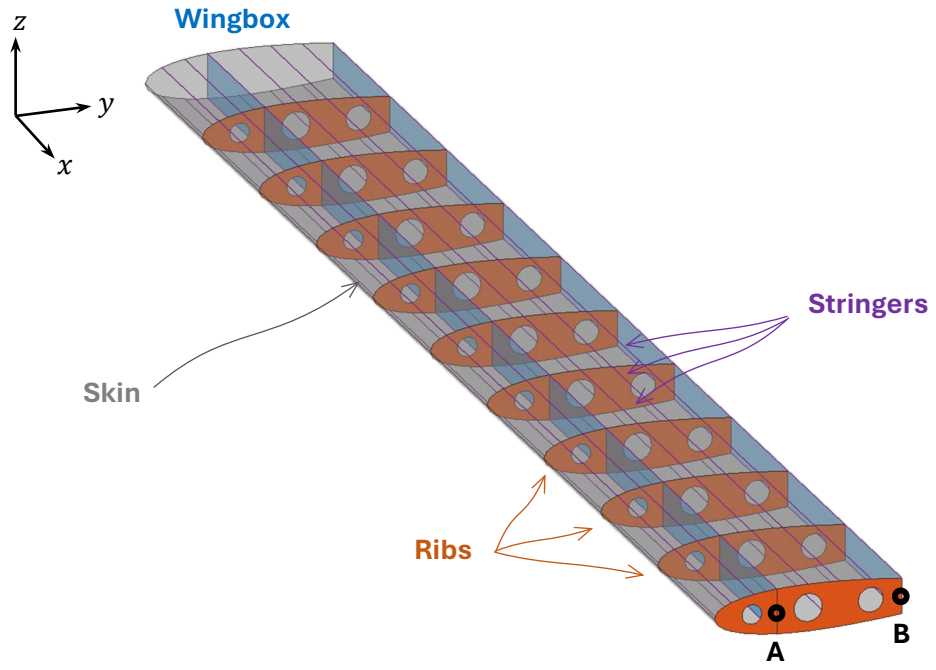
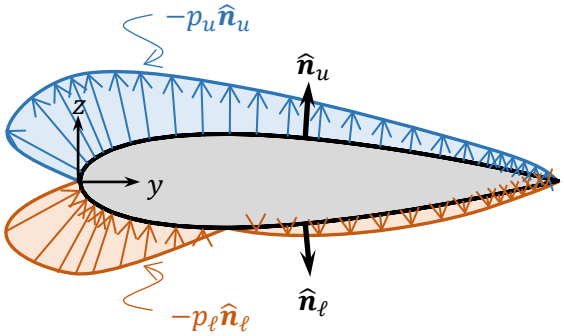


Figure 2. Wing structure.

1) The first goal is to analyze the effects of adding these different elements to the overall structure. To do so:
   a) Build the stiffness and mass matrices for the **wingbox** modelled with shell elements (use the **`Tn_wb`** and **`Tm_wb`** variables as nodal and material connectivities matrices).
   b) Build the stiffness and mass matrices for the **stringers** modelled with beam elements of circular section with 15 mm diameter, considering $k_{y'} = k_{z'} = 5/6$ and $k_t = 1$ (use the **`Tn_st`** and **`Tm_st`** variables as nodal and material connectivities matrices).
   c) Build the stiffness and mass matrices for the **ribs** modelled with shell elements (use the **`Tn_rb`** and **`Tm_rb`** variables as nodal and material connectivities matrices).
   d) Build the stiffness and mass matrices for the **skin** modelled with shell elements (use the **`Tn_sk`** and **`Tm_sk`** variables as nodal and material connectivities matrices).
   e) Repeat the **bending** and **torsion** tests of part II and obtain the same results for the cases considering *all* or *some part* of the structural elements combined.

   Hint: To study the effects of each element separately, the corresponding mass and stiffness matrices can be switched on/off. *Example*: multiplying by zero (or a very small constant) the stiffness and mass matrices for the skin elements virtually removes them from the structure.

2) A wind tunnel test is now performed so the wing is subjected to its own **weight** and a **pressure** load distributed over the upper and lower skin surfaces, according to:



$$p_u(x,y) = -\alpha P(x)\left[\left(1-\frac{y}{c}\right)^4 + \sqrt{1-\frac{y}{c}}\right]$$

$$p_\ell(x,y) = \alpha P(x)\left[\left(1-\frac{y}{c}\right)^4 - \frac{1}{4}\sqrt{1-\frac{y}{c}}\right]$$

$$P(x) = \begin{cases} p_\infty\left(0.5 + 4.8\dfrac{x}{b} - 11.2\left(\dfrac{x}{b}\right)^2\right) & x \in [0, 0.25b] \\ p_\infty\left(1.2 - 0.8\dfrac{x}{b}\right) & x \in [0.25b, 0.75b] \\ p_\infty\left(-2.4 + 8.8\dfrac{x}{b} - 6.4\left(\dfrac{x}{b}\right)^2\right) & x \in [0.75b, b] \end{cases}$$

with $\alpha = 20°$ being the angle of attack (expressed in radians), and $p_\infty = 1.5 \times 10^5$ Pa.

Hint: To define the external pressure force distribution, matrices stored in the **`n_u`** and **`n_l`** variables contain the components of the unit normal vectors for the upper and lower skin surfaces, respectively, on the first three columns, and the indices for the corresponding nodes on the fourth column.

In this scenario:

a) Solve the **static** problem and obtain the average vertical deflection $\bar{u}_z$ and twist angle $\bar{\theta}_x$ distributions along the wingspan (use formulas in page 3).
b) Obtain and plot the Von Mises stress distribution in the wingbox, ribs and skin elements separately.

3) Develop and apply a **model-order reduction** strategy based on a modal projection to solve the wind tunnel test problem with a reduced set of natural modes, comparing the average vertical deflection and twist angle distributions, as well as the Von Mises stress distribution obtained in each case with those of the complete problem (solution of point III.2). Justify which modes are more relevant for this case and assess the **accuracy** and speed-up gained in terms of **computation time**.

Note: The mesh variables contained in **`wing.mat`** file are: **`xn`** (nodal coordinates matrix), **`Tn_st`** and **`Tm_st`** (nodal and material connectivities matrices for the stringers), **`Tn_sk`** and **`Tm_sk`** (nodal and material connectivities matrices for the skin), **`Tn_rb`** and **`Tm_rb`** (nodal and material connectivities matrices for the ribs), and **`Tn_wb`** and **`Tm_wb`** (nodal and material connectivities matrices for the winbgox). In the latter case, material index 1 refers to the front spar, 2 refers to the rear spar, and 3 to the upper and lower surfaces of the wingbox. In addition, the **`indRoot`**, **`indPointA`**, **`indPointB`**, **`indSpar1`**, and **`indSpar2`** variables are also included in this case (see part II for details). The script file **`MAIN_wing.m`** can be used as template to develop the code to get the requested results.

Important: Initialize the global mass and stiffness matrices using the **`sparse`** function instead of the **`zeros`** function. This will allow you to use the **`eigs`** function for the modal analysis obtaining just the needed number of modes (and speeding up the computation).